

MARIAM GAMAL

Computer Science Engineer | Software Engineering & AI

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SUMMARY

Computer Science Engineering student with hands on experience in full stack web development and applied AI. Built multiple end to end applications using Python, JavaScript/TypeScript, and modern web frameworks, and developed Retrieval Augmented Generation (RAG) and deep learning systems using PyTorch, LangChain, and FAISS. Strong foundation in data structures, algorithms, debugging, REST API design, and software engineering best practices.

EDUCATION

Egypt-Japan University of Science and Technology (E-JUST) 2024 – Present
B.Sc. in Computer Science and Engineering

El Zahraa International Schools - American Department 2020 – 2023
High School Diploma, American Division

SKILLS

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, C++, C, HTML, CSS
- **Web & Frameworks:** React.js, Next.js, Node.js, Express.js, Flask, Bootstrap, Tailwind CSS
- **Databases:** MySQL, MongoDB, PostgreSQL (Prisma ORM)
- **AI / Machine Learning:** PyTorch, LangChain, FAISS, Hugging Face Embeddings, Sentence Transformers, PyPDFLoader, Prompt Engineering
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, REST API Design, Debugging & Testing
- **Tools:** Git/GitHub, VS Code, Android Studio

Soft Skills & Languages

- **Soft Skills:** Problem Solving, Team Collaboration, Communication, Adaptability, Continuous Learning, Time Management, Attention to Detail
- **Languages:** Arabic (Native), English (Fluent - EF SET Certified)

PROJECTS

Full-Stack Learning Management System (LMS) 2026
Next.js 15 | Express.js | TypeScript | Prisma | PostgreSQL (Neon) | Vercel

- Developed a production grade monorepo LMS with a Next.js 15 App Router frontend and an Express + TypeScript + Prisma backend, deployed on Vercel with a live Neon PostgreSQL database.
- Implemented 5 specialized user roles (Admin, Doctor, TA, Student, Support Agent), full Arabic/English RTL/LTR switching, and a custom design system with glassmorphic UI elements and micro animations.

Aura - Customer Ordering System 2026
Flask | Python | Full-Stack Web

- Built a full-stack ordering platform (dynamic menu, real time order tracking, admin dashboard) with RESTful Flask routes following MVC and OOP principles.

AI Knowledge & Reasoning Engine 2025
Python | LangChain | FAISS | Hugging Face | PyTorch

- Built a RAG pipeline over multi-document PDF corpora using PyPDFLoader, Hugging Face Sentence Transformers, and FAISS for low latency similarity search, orchestrated with LangChain.
- Applied prompt engineering and iterative testing to improve answer grounding and reduce hallucinated responses.

WORK EXPERIENCE

Information Technology Institute (ITI) Aug 2026 - Present
Software Testing

- Conducting comprehensive software testing procedures to ensure application quality and functionality across multiple projects.
- Collaborating with development teams to identify and resolve defects throughout the software development lifecycle, implementing automated testing solutions.

Route Academy - IT Training Center June 2026 - Present
Front-End Development Diploma

- Undertaking hands-on training in HTML, CSS, React.js, Bootstrap, Tailwind, TypeScript, Sass, jQuery, NPM and modern Java Script (ES6, DOM, BOM, AJAX, Fetch API, OOP).

SyntecxHub Company
UI/UX Design Internship

Jun 3 – Jul 3, 2026

- Designed and developed 4 high-fidelity UI/UX projects during a one month internship, applying user centered design principles and pixel perfect front-end implementation; Created wireframes and prototypes reducing development iteration time by 25% through early user feedback integration.

Decode Labs Company
AI Internship

Jun 1 – Jul 1, 2026

- Worked on applied AI projects during a one month internship, strengthening practical skills in AI development and implementation.

Tips Hindawi Company
Generative AI Internship Program (126 hrs)

Dec 2025 – Feb 2026

- Completed an intensive program on generative AI concepts, including LLMs and applied prompt engineering, through hands on implementation exercises.

Saham Al Shamal Eng'g Consultants Co. (SASEC)
Online Training IT Intern

Jul – Aug 2025

- Completed technical training on engineering consultancy processes and systems.

Alexandria Mineral Oils Company (AMOC)
Industrial IT Intern

Sep 2024

- Monitored industrial control systems processing 1000+ data points daily, identifying and resolving 15+ operational issues reducing downtime by 20%; performed data analysis on production metrics, creating automated reports saving 10+ hours weekly in manual data processing.

CERTIFICATIONS

- Elements of AI Certificate — University of Helsinki & MinnaLearn
- EF SET C2 Proficient (77/100) Certificate
- AMOC Industrial IT Training Certificate
- SASEC CS Internship Certificate
- Tips Hindawi Generative AI Internship Certificate
- Decode Labs AI Internship Certificate
- Syntecx Hub UI/UX Design Internship Certificate